GM FINANCIAL CTO ORGANIZATION

DELIVERY TWIN

AI Delivery Intelligence Platform

From idea to outcome, continuously.



Hackathon Team 35

Shape → Sense → Forecast → Act

A continuous decision loop that connects commitment, execution signals, prediction, and recovery.

EXECUTIVE SUMMARY

Projects fail in the gap between commitment and execution

Point-in-time estimates age quickly; delivery conditions change every sprint.

THE BLIND SPOT

- Scope expands after approval
- Critical skills are diverted to production support
- Infrastructure and change requests stall
- Green status persists after delivery reality changes

DELIVERY TWIN CLOSES THE LOOP

SHAPE

Create an explainable commitment

SENSE

Continuously read delivery conditions

FORECAST

Quantify likely outcomes

ACT

Select and synchronize the recovery plan

Outcome: decisions stay aligned to current delivery reality, not the original assumption set.

1. SHAPE → 2. SENSE → 3. FORECAST → 4. ACT

OPERATING MODEL

Four stages, one continuous delivery decision loop

The executive experience stays simple while the underlying capabilities remain rich.

1. SHAPE

What will it take?

- Incomplete intake
- ROM + assumptions
- Capacity + baseline

2. SENSE

What changed?

- Labor + PTO
- Velocity + scope
- Operational blockers

3. FORECAST

Where are we heading?

- Monte Carlo range
- Confidence score
- Portfolio ripple

4. ACT

What should we change?

- What-if scenarios
- Human approval
- Closed-loop sync

Every approved action creates the new baseline for the next sensing cycle.

1. SHAPE → 2. SENSE → 3. FORECAST → 4. ACT

Turn incomplete demand into an explainable commitment

Combine intake clarification, structural similarity, effort ranges, skills, and availability before locking the plan.



Incomplete intake

AI generates RAM-style clarification questions for architecture, data, security, and reuse.

6D delivery fingerprint

Match structure, complexity, systems, roles, dependencies, and delivery pattern.

ROM with confidence

P50 effort: 4,450 hours
P80 effort: 4,950 hours
Explicit 14% integration buffer.

Feasible commitment

7.5 of 8.0 FTE available
March 31 target
84% initial confidence.

Decision gate: approve the baseline only when assumptions, capacity, and confidence are visible.

1. SHAPE → 2. SENSE → 3. FORECAST → 4. ACT

Continuously detect changes to the delivery conditions

Connect workforce reality, sprint execution, and operational queues to the approved baseline.

02 SENSE

UKG Pro WFM

Actual time, PTO, shifts

Lead engineer loses 16 hours/week to P1 support plus 40 hours PTO.

Oracle HR / HCM

Active headcount and skills

Single-threaded data engineering dependency becomes visible.

Azure DevOps

Velocity, scope, backlog

14 unplanned stories add 64 points for legacy adapters.

ServiceNow

Incidents, changes, blockers

Kafka production-mirror change remains in CAB review for nine days.

ZERO-PROMPT FINDING

The plan still appears green, but the combined signals show declining effective capacity, rising scope, and an unresolved environment dependency.

1. SHAPE → 2. SENSE → 3. FORECAST → 4. ACT

Translate live signals into a probability-based delivery outlook

Deterministic simulation produces the numbers; AI explains the drivers and business impact.

03 FORECAST

CONFIDENCE	FORECAST	SIMULATIONS
38%	**+19 days**	**10,000**
Probability of Sprint 4 delivery	Moves completion into Sprint 5	Propagating labor and scope variance

PORTFOLIO RIPPLE EFFECT

A Sprint 5 slip breaches the shared API gateway milestone required by the Q2 Auto-Pay Modernization value stream.

Forecast output includes completion range, confidence, financial exposure, impacted milestones, and explainable drivers.



AUTONOMOUS CORRELATION

Workforce drain + scope growth + CAB delay = multi-system collision.

1. SHAPE → 2. SENSE → **3. FORECAST** → 4. ACT

Choose the recovery path, approve the trade-off, and re-plan

What-if simulation compares date, cost, scope, portfolio impact, and confidence.

04 ACT

OPTION A

Contractors

+$118K

Onboarding prevents Sprint 4 recovery

Modeled confidence 68%

OPTION B

Squad reallocation

$0

Delays Audit Reporting by 4 weeks

Modeled confidence 79%

OPTION C

AI PICK

Phase + unblock

$0

Defers cold archive, fast-tracks CAB, shields lead engineer

Modeled confidence 92%



APPROVE & SYNC

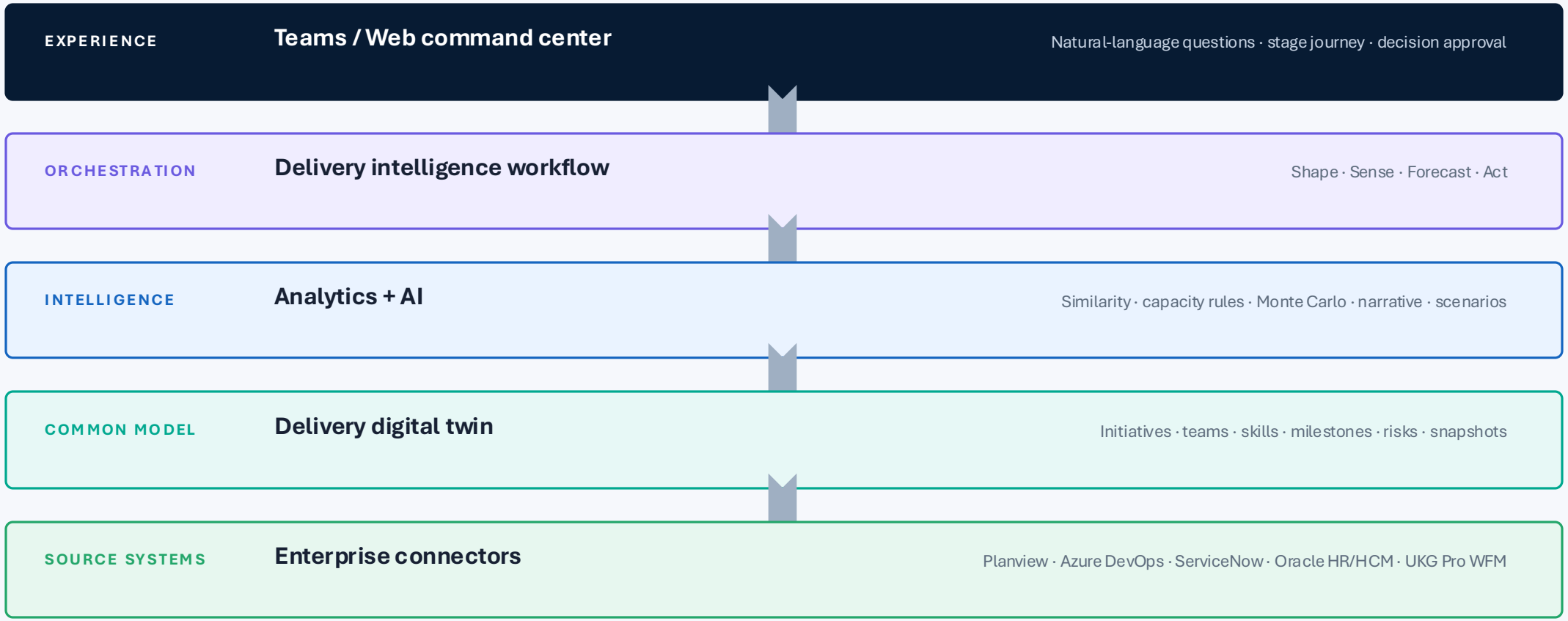
Lock the revised milestone, update sprint scope, route the change request, and protect the critical skill allocation.

1. SHAPE → 2. SENSE → 3. FORECAST → 4. ACT

SOLUTION ARCHITECTURE

AI explains. Deterministic engines calculate. Humans approve.

The hackathon prototype can use synthetic data and simulated write-back while preserving the production design.



Prototype guardrail: all connectors and write-back actions are simulated unless explicitly enabled and approved.

DEMO STORYLINE

Build 20%. Tell 100%.

One initiative moves through the full lifecycle in a five-minute executive narrative.

1. **SHAPE** — Messy intake becomes an explainable ROM and approved baseline.
2. **SENSE** — Fast-forward to Week 5; labor, scope, velocity, and CAB signals diverge.
3. **FORECAST** — The system predicts a 19-day slip and exposes the downstream milestone impact.
4. **ACT** — Leadership compares three recovery paths and approves the 92%-confidence plan.

CLOSING LINE **“We did not just estimate the project. We continuously found the plan most likely to succeed.”**

WHY THIS CAN WIN

It turns fragmented delivery data into an executive decision advantage

The differentiator is not another dashboard. It is proactive discovery plus a credible recovery decision.

MEMORABLE

A clear four-stage story judges can repeat.

DIFFERENTIATED

Zero-prompt cross-system collision detection.

TRUSTWORTHY

Deterministic calculations with explainable AI narratives.

ACTIONABLE

Scenario trade-offs plus human-approved re-planning.

THE ASK

Align the team on the four-stage story, build the decisive demo moments, and treat enterprise integrations as the production path.

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